

# Escriba

The universal translator into the language of AI

Complete manual — users & administrators

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[github.com/diegoparras/escriba](https://github.com/diegoparras/escriba)

# Part 1 — For users

## What is Escriba?

Escriba converts almost any file into clean Markdown — plain text with formatting, ready to copy, save or feed to an AI. It handles documents, spreadsheets, slides, images, audio, video and YouTube.

## Getting started

- Enter the password and sign in (your level: HUMANO, ANGEL or DIOS).
- Drag files onto the centre area, or click “Select files”.
- Click “Convert all”. A progress bar appears.
- Open the result and use Preview / Markdown / Split. Copy it or download the .md (or a .zip for several files).

## Key features

- Audio & video are transcribed locally with Whisper; pick the language under Advanced options.
- Paste a YouTube link to get its transcript.
- Text in images is OCR'd automatically; scanned PDFs are detected, OCR'd and auto-rotated on the fly.
- AI is optional (OpenAI, Gemini, OpenRouter) and defaults to “No AI”. Models are listed automatically.
- PII anonymization for LLMs: a local engine (NER + invoice layout + validated detectors + your own rules) with five output modes — typed, anonymous, reversible pseudonymization (token → send to the LLM → re-hydrate), partial masking (••••-3456) and stable hashing (same data → same pseudonym across documents).
- Visual redaction: download your PDF or scanned image with the PII blacked out on the page — true redaction: the text and the pixels underneath are removed from the file.
- Queue: convert everything or only the selected files.
- Light/dark theme and an interface available in 7 languages.
- Export beyond Markdown: one unified menu turns the result into Word (.docx), ODT, EPUB, HTML, LaTeX, reStructuredText or structured XML (DocBook, JATS, TEI, OPML), via Pandoc — no AI involved.
- LLM prep panel on every conversion: token count, tokens and cost saved by anonymization, a live per-model cost estimate (OpenRouter pricing), context-window fit, one-click RAG chunking and a prompt-injection detector — all local.
- Advanced PDF extraction (optional): the OpenDataLoader engine reconstructs complex layouts with better reading order and heading hierarchy, falling back to the default extractor automatically.

## Part 2 — Roles & access levels

Login is required. Each level has its own password and limits, all configurable via environment variables.

| Capability                | HUMANO | ANGEL  | DIOS      |
|---------------------------|--------|--------|-----------|
| Upload & convert files    | Yes    | Yes    | Yes       |
| Convert from a public URL | No     | Yes    | Yes       |
| Internal URL / local path | No     | No     | Yes       |
| Audio / video / ZIP       | No     | Yes    | Yes       |
| YouTube transcripts       | Yes    | Yes    | Yes       |
| OCR (forced / automatic)  | No     | Yes    | Yes       |
| Use server AI keys        | No     | Yes    | Yes       |
| Max file size             | 25 MB  | 100 MB | unlimited |
| Files per batch           | 3      | 10     | unlimited |
| Rate limit (req/min)      | 15     | 60     | unlimited |

## Part 3 — For administrators

### Architecture

Escriba runs in a single, self-contained Docker container that bundles FastAPI + Markdown, ffmpeg, Tesseract + OCRmyPDF, faster-whisper and an embedded Redis. It runs as a non-root user with a healthcheck and automatic CPU-core detection.

### Deployment

Run the prebuilt image (no build needed):

```
docker run -d --name escriba -p 8000:8000 \
  -e SECRET_KEY="<long-random-key>" \
  -e GOD_PASSWORD="<your-password>" \
  ghcr.io/diegoparras/escriba:latest
```

Or with Docker Compose:

```
git clone https://github.com/diegoparras/escriba.git
cd escriba
cp .env.example .env
docker compose up -d --build
```

EasyPanel/Portainer/Dokploy: create an App from the Docker image ghcr.io/diegoparras/escriba:latest, add the environment variables, map the domain to container port 8000 with HTTPS, and deploy.

## Environment variables

| Variable                         | Default   | Description                                |
|----------------------------------|-----------|--------------------------------------------|
| SECRET_KEY                       | random    | Session signing key. Set it in production. |
| GOD/ANGEL/HUMAN_PASSWORD         | —         | Password for each access level.            |
| HUMAN_OPEN                       | false     | Allow the HUMANO level without login.      |
| WEB_CONCURRENCY                  | auto      | Parallel workers (auto = CPU cores).       |
| MAX_UPLOAD_MB                    | 100       | Maximum upload size (except DIOS).         |
| WHISPER_MODEL                    | base      | Transcription model (tiny ... large-v3).   |
| MAX_MEDIA_MINUTES                | 120       | Audio/video length cap (0 = unlimited).    |
| OPENAI/OPENROUTER/GOOGLE_API_KEY | —         | Server-side AI keys (fallback).            |
| API_TOKEN / API_TOKEN_ROLE       | — / angel | Token for automation and its role.         |
| EMBEDDED_REDIS                   | true      | Embedded Redis for shared rate limiting.   |
| ENABLE_DOCS                      | false     | Expose Swagger at /api/docs.               |

## Server-side AI keys

- If set, users don't need to paste their own key. Only DIOS and ANGEL use them.
- The cost is on you and is shared. Leave them empty so each user brings their own.
- A user-supplied key always takes priority over the server key.

## Transcription (Whisper)

- Set the model with WHISPER\_MODEL (larger = more accurate, slower).
- It runs on CPU. MAX\_MEDIA\_MINUTES caps the duration (DIOS has no cap).

## API token (automation)

For n8n or scripts, set API\_TOKEN and send it as a header:

```
curl -H "X-API-Key: YOUR_TOKEN" \
  -F "file=@document.pdf" \
  https://your-domain/api/convert
```

## Security

- Authentication required; three roles with distinct permissions.
- Anti-SSRF (internal IPs/redirects blocked); local paths only for DIOS.
- XSS sanitization (DOMPurify) and security headers (CSP).
- Non-root container with no-new-privileges; OCR/Whisper run without a shell.
- Per-role rate limiting; the API key is never stored on the server.

## Privacy

Uploaded files are processed in a temporary file and deleted as soon as the conversion finishes (success or error). There is no database and no uploads folder; the result is only sent to your browser.

## Part 4 — API reference

| Endpoint          | Method | Description                                                                           |
|-------------------|--------|---------------------------------------------------------------------------------------|
| /api/login        | POST   | Log in; sets a session cookie.                                                        |
| /api/me           | GET    | Current role and capabilities.                                                        |
| /api/convert      | POST   | Convert a file or URL to Markdown.                                                    |
| /api/models       | POST   | List the provider's models.                                                           |
| /api/stats        | GET    | Server stats (by role).                                                               |
| /api/export       | POST   | Convert Markdown to docx, odt, epub, html, latex, rst or XML (DocBook/JATS/TEI/OPML). |
| /api/compact      | POST   | Whitespace-stripped Markdown to save tokens.                                          |
| /api/chunk        | POST   | Token-bounded RAG chunks (.jsonl).                                                    |
| /api/model_prices | GET    | Live model pricing and context windows (OpenRouter, cached).                          |

POST /api/convert (multipart/form-data): file or url, plus optional lang, ocr, llm\_provider, llm\_api\_key, llm\_model.

## Part 5 — Troubleshooting

- A long video won't transcribe: it may exceed MAX\_MEDIA\_MINUTES — raise the cap or use DIOS.
- A YouTube video fails: it may have no captions, or YouTube blocks your server IP — paste your YouTube cookies in Settings, or set YT\_PROXY/YT\_COOKIES.
- Error 413 on upload: raise MAX\_UPLOAD\_MB and your reverse proxy's body limit.
- A PDF came out empty: it was scanned without OCR for your role — use ANGEL/DIOS or tick “Force OCR”.
- Forgot the DIOS password: if unset, a random one is printed in the container logs at startup.

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